

Multiple Choice (10 marks)

1. In MD-5 the length of the message digest is
 - (a) 160
 - (b) 128
 - (c) 64
 - (d) 54
2. Encryption and decryption provide secrecy, or confidentiality, but not
 - (a) Authentication
 - (b) Integrity
 - (c) Privacy
 - (d) All of the above
3. Why would a ping sweep be used?
 - (a) To identify live systems
 - (b) To locate live systems
 - (c) To identify open ports
 - (d) To locate firewalls
4. Which of the following deals with network intrusion detection and real-time traffic analysis?
 - (a) John the Ripper
 - (b) L₀ phtCrack
 - (c) Snort
 - (d) Nessus
5. How is IP address spoofing detected?
 - (a) Installing and configuring a IDS that can read the IP header
 - (b) Comparing the TTL values of the actual and spoofed addresses
 - (c) Implementing a firewall to the network
 - (d) Identify all TCP sessions that are initiated but does not complete successfully

True / False (10 marks)

Answer True or False

6. True or False: Only biometric methods are considered effective forms of authentication.
7. True or False: WEP is considered the least secure encryption standard among wireless security protocols.
8. True or False: SHA-1 generates a message digest that is 512 bits long.
9. True or False: A crypter is primarily used to create malicious software that disguises its true nature by altering its code structure.
10. True or False: ECB, CFB, and CBC are all valid block cipher operating modes used in encryption.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the number of rotations in a circularly sorted array.
12. Write a function to compute the value of $nCr \bmod p$.

End of Paper